EUROPEAN COMMISSION
Executive Agency for Small and Medium-sized Enterprises
H2020 SME

ANNEX 1 (part A)

SME instrument phase 2

NUMBER — 823392 — MOGU floor

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1.1. The project summary

Project Number [1]	823392	Project Acronym [2]	MOGU floor

One form per project

General information

Project title [3]	Natural-Grown Flooring for Circular Buildings
Starting date [4]	01/10/2018
Duration in months [5]	24
Call (part) identifier [6]	H2020-SMEInst-2018-2020-2
Topic	EIC-SMEInst-2018-2020 SME instrument
Fixed EC Keywords	Design innovation, Architecture, smart buildings, smart cities, urban engineering, Bioproducts (products that are manufactured using biological material as feedstock) biomaterials, bioplastics, biofuels, bioderived bulk and fine chemicals, bio-derived novel materials
Free keywords	Circular economy, Bio-economy, Mycology, Green building

Abstract [7]

The green building market is anticipated to be among the fastest growing industries worldwide. Driven by citizen and building owner demand for sustainability and operational cost reduction, the green building market is, indeed, doubling every three years. To satisfy the market demand for sustainable and functional building solutions we developed MOGU floor, a disruptive product for the resilient flooring market. MOGU floor is a 100% biobased tile made from agricultural or industrial residual biomass. It outperforms the competing solutions already available on the market in terms of safety, fire resistance, thermal insulation, shock and noise absorption. The entire lifecycle of MOGU floor, including production, installation, maintenance and disposal, represents a new circular value chain for the building market, highly sustainable and profitable. Moreover MOGU floor has a distinctive and attractive design and tactile feeling that make it a perfect solution for luxury living environments at a target price typical of inferior market segments. To take advantage of all the unique selling points of MOGU floor we developed a market strategy aiming at quickly gain international market share and become in the mid-term a reference player for innovation, sustainability and design in the flooring market. To achieve such result, we elaborated a number of actions in collaboration with highly influential stakeholder of the building market to increase customer awareness and demand for MOGU floor. Now we are applying to the SME instrument ph2 as part of our fundraising strategy to finalize the development of the first MOGU floor collection and accelerate its market uptake. During the 24 months workplan we elaborated we are going to certify MOGU floor with strategic green labels, upscale our production capacity, reinforce our IPR and market positioning and start the b2b collaborations to promote, commercialize and distribute MOGU floor.

1.2. List of Beneficiaries

Project Number [1]	823392	Project Acronym [2]	MOGU floor

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	MOGU SRL	MOGU	Italy	1	24

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1.3.1. WT1 List of work packages

WP Number [9]	WP Title	Lead beneficiary[10]	Person-months [11]	Start month [12]	End month [13]
WP1	Designing of MOGU floor collection	1 - MOGU		1	4
WP2	Product Manufacturing Optimization	1 - MOGU		3	6
WP3	Production upscale and Digitalisation	1 - MOGU		5	24
WP4	Certifications	1 - MOGU		5	24
WP5	Gear up for commercialisation	1 - MOGU		1	24
WP6	Management	1 - MOGU		1	24
WP7	Ethics requirements	1 - MOGU		1	24
		Total			

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Report on the selected certification for MOGU floor	WP1	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	4
D1.2	MOGU floor technical sheet	WP1	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	4
D2.1	Manufacturing protocol and guidelines of the first MOGU floor collection	WP2	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D3.1	Design of the upscaled production plant	WP3	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D3.2	Operational manual of the production plant	WP3	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D3.3	Operational manual of the monitoring system	WP3	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	22
D3.4	Prototypes samples produced in the upscaled plant	WP3	1 - MOGU	Demonstrator	Public	18
D4.1	Report on MOGU floor certifications	WP4	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.2	Report on the manufacturing process certifications	WP4	1 - MOGU	Report	Confidential, only for members of the consortium	24

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					(including the Commission Services)	
D5.1	New MOGU website	WP5	1 - MOGU	Websites, patents filling, etc.	Public	3
D5.2	Communication Plan	WP5	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D5.3	Market Review	WP5	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D5.4	Report on status of patents or other IPR tools	WP5	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D5.5	Business Plan for the commercialisation phase	WP5	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D6.1	Internal project management tool	WP6	1 - MOGU	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	1
D6.2	Compilation of Quarterly Reports to the Management Board for Year One	WP6	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D6.3	Compilation of Quarterly Reports to the Management Board for Year Two	WP6	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	H - Requirement No. 1	WP7	1 - MOGU	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2
D7.2	POPD - Requirement No. 2	WP7	1 - MOGU	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2

Timing of the work packages and their components (include a Gantt chart or similar).

ACTIVITY	MONTHS																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
WP1 Designing of MOGU floor collection				M																				
Selection of product certification				D																				
Definition of the technical specifications				D																				
Definition of the aesthetic design																								
WP2 Product Manufacturing Optimization						M																		
Optimization of MOGU growing phase																								
Optimization of post-production						D																		
WP3 Production upscale and digitalisation										M								M						M
Upscaled plant design										D														
Building and commissioning of the new plant																		D						
Increase process automation and digitalization																						D		
Demonstration with early adopters																								
WP4 Certifications													M					M						M
Achievement of CE mark																								
Achievement of additional green certifications																		D						
Achievement of process certifications																								D
WP5 Gear up for commercialisation																		M						M
echnology Watch and Knowledge Management																								
Market and Business Planning									D															D
Preparation for commercialisation																		D						
Communication, dissemination and marketing			D			D																		
WP6 Management																								
Monitoring and reporting	D											D												D
Risk Management																								
Innovation Management								D								D								D

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	1 - MOGU
Work package title	Designing of MOGU floor collection		
Start month	1	**End month**	4

Objectives

1) Define the technical and aesthetic specifications of the first MOGU floor collection in co-creation with clients and end-users to meet market demand in compliance with industrial standards,
2) Select which product certifications will be applied to MOGU floor in order to maximise the value for clients and end-users, according with the commercialization strategy

Description of work and role of partners

WP1 - Designing of MOGU floor collection [Months: 1-4]
MOGU
Task 1.1: Selection of product certifications
A wide range of "green" product certifications exist at international level. The selection of the most valuable ones for MOGU floor is instrumental to reinforce our USPs and internationalization strategy. The certifications selected in this task will be implemented in WP4. The first certification to be achieved is the CE mark, which is needed to obtain a marketable product, as specified in the Construction Products Regulation (CPR) 305/2011/EU. In addition to the CE mark, we will select further certifications to apply for based on: regional relevance, competitors certifications, value, granting procedure and MOGU floor characteristics. From our analysis MOGU floor is suitable for certification such as BREEAM, DGNB, Indoor airPLUS, Nordic Swan and others.

Task 1.2: Definition of the technical specifications
The CE mark sets the basic specification to comply with and therefore represent the "must have" features of MOGU floor. We have already tested in our lab MOGU floor for CE mark compliance, according with the ISO 14041:2004/ AC:2006 "Resilient, textile and laminate floor coverings. Essential characteristics", and the all results obtained are well beyond the minimum threshold.
To maximise the value of MOGU floor we will also define the "nice-to-have" features. We have already verified the performance of MOGU floor for several parameters according to ISO 16000-9:2006, which defines construction materials behaviour for indoor air-quality and ISO 10874:2009 that classifies the type of flooring.
The result of this task will be the elaboration of the MOGU floor technical sheet, which will describe the technical characteristics of our product and that will accompany each sell. On the technical sheet will be included also the graphic symbols prepared by the Technical Committee CEN/TC 134 "Resilient, textile and laminate floor coverings" for helping end-user understanding of MOGU floor characteristics.

Task 1.3: Definition of the aesthetic design
In this task, we will investigate stakeholders and end users demand in terms of design and aesthetic characteristics. MOGU floor has already gained wide appreciation for its unique natural look and feel and we will work closely to the Commercialization Team in WP5, to identify which shapes, colour and coating solutions are capable of offering the best physical, mechanical, visual and tactile characteristics, in order to identify the best designs to be placed on the market for meeting consumer panel results. The consumer panel will be organized by M3, with the voluntary support of stakeholders such EXT (see letter of support), and will be set in Milan. Architects, designers, suppliers and end users will be invited and open to approximately 60 people. The objective is to select, in a pool of 15 prototypes.

Participation per Partner

Partner number and short name	WP1 effort
1 - MOGU	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type [15]	Dissemination level [16]	Due Date (in months)[17]
D1.1	Report on the selected certification for MOGU floor	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	4
D1.2	MOGU floor technical sheet	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	4

Description of deliverables

D1.1 Report on the selected certification for MOGU floor [M4]
D1.2 MOGU floor technical sheet [M4]

D1.1 : Report on the selected certification for MOGU floor [4]

Report on the selected certification for MOGU floor

D1.2 : MOGU floor technical sheet [4]

MOGU floor technical sheet

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Design of the first MOGU floor collection	1 - MOGU	4	The first portfolio of MOGU floor will be presented: online, press release, events

Work package number [9]	WP2	**Lead beneficiary** [10]	1 - MOGU
Work package title	Product Manufacturing Optimization		
Start month	3	**End month**	6

Objectives

1) Optimize the growing phase of MOGU technology to properly meet the product specification,
2)Optimize the drying and finishing phase of MOGU technology to properly meet the product specification

Description of work and role of partners

WP2 - Product Manufacturing Optimization [Months: 3-6]
MOGU
Task 2.1 Optimization of MOGU growing phase
In this WP, our consolidated protocols will be further improved and optimized to meet the final design of MOGU floor and extra care will be given to maintain the obtained results in the scaling-up process. The optimal growing parameters has already been defined, so the optimization of the plant will study conditions such as the best space distribution and heat transfer to increase efficiency and quality.

Task 2.2 Optimization of post-production
In partnership with ORINEO, soft bio-based resins have been tested for MOGU Floor finishing, showing very interesting results: the tiles show excellent thermo-mechanical characteristics, and the resin ensure durability, water resistance and so on. Also, the cut edge precision and dimensional standardization are improved, speeding up the installation by hours. Through this task MOGU Floor will be further optimized setting up a process capable to achieve a homogeneous quality product, optimize logistic and plant layout.

Participation per Partner

Partner number and short name	**WP2 effort**
1 - MOGU	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type** [15]	**Dissemination level** [16]	**Due Date (in months)**[17]
D2.1	Manufacturing protocol and guidelines of the first MOGU floor collection	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

D2.1 Manufacturing protocol and guidelines of the first MOGU floor collection [M6]

D2.1 : Manufacturing protocol and guidelines of the first MOGU floor collection [6]
Manufacturing protocol and guidelines of the first MOGU floor collection

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS2	Manufacturing protocol of MOGU floor	1 - MOGU	6	The full production protocol and conditions will be described in D2.1

Work package number [9]	WP3	**Lead beneficiary** [10]	1 - MOGU
Work package title	Production upscale and Digitalisation		
Start month	5	**End month**	24

Objectives

1) Design and build the up-scaled MOGU production plant,
2) Develop a monitoring and control system for the efficient running of MOGU floor production,
3) Automating and digitalizing the production process

Description of work and role of partners

WP3 - Production upscale and Digitalisation [Months: 5-24]
MOGU
Task 3.1 Plant scale-up design
We will scale up to maximum production of the current plant placed in Inarzo (Varese), from 50 to 600 m2/day, applying the principles of industry 4.0 and develop a highly efficient production system. An efficient monitoring system will be set up in order to make the process more efficient, easy to control and, eventually, adjust in course.
Out team cover the expertise needed to successfully accomplish the designing of the upscaled plant and we will take advantage of the support of a subcontractor (described in section 4) for defining the control system.

Task 3.2 Building and commissioning of the upscaled plant
Once the upscaled design is finished, we will place the purchases and then start the construction phase. Our team will select the equipment and take care of the customization needed for such tailor-made process. We will work also in collaboration with the subcontractor (described in section 4) for the installation of the monitoring and control system. The construction of the plant will be performed in modules in order to allow the preproduction and testing of equipment before the full scale plant will be finished.

Task 3.3 Increase process automation and digitalization
The manufacturing operations need to be flexible and adjustable, to meet the normally variable characteristics of raw materials. With the support of the subcontractor (described in section 4), we will develop an advanced monitoring system based on photonics (NIR and/or image recognition) to control the critical process parameters along the production line. The data acquired will be elaborated applying Artificial Intelligence algorithms and a predictive model will be developed. The model will provide information to adapt the process conditions to fluctuations and thanks to the implementation of Machine Learning will be able to adapt to future productions.

Task 3.4 Demonstration with early adopters
Once the upscaled facility is fully operational we will validate the selected portfolio of products in WP1, in collaboration with stakeholders such as ATI Projects, ExT, Vaxana and Arup (see letter of support) that will test MOGU floor in their installations as early adopters. They will receive MOGU floor samples, together with their technical sheet, in order to evaluate their installation, performance and user appreciation in some demonstrative installation. The same approach will be followed with additional stakeholder and will be also used for promotion.

Participation per Partner

Partner number and short name	WP3 effort
1 - MOGU	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type [15]	Dissemination level [16]	Due Date (in months)[17]
D3.1	Design of the upscaled production plant	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D3.2	Operational manual of the production plant	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D3.3	Operational manual of the monitoring system	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	22
D3.4	Prototypes samples produced in the upscaled plant	1 - MOGU	Demonstrator	Public	18

Description of deliverables

D3.1 Report on the design of the upscaled production plant [M10]
D3.2 Operational manual of the production plant [M18]
D3.3 Operational manual of the monitoring system [M20]
D3.4 Prototypes samples produced in the upscaled plant [M18]

D3.1 : Design of the upscaled production plant [10]
Design of the upscaled production plant

D3.2 : Operational manual of the production plant [18]
Operational manual of the production plant

D3.3 : Operational manual of the monitoring system [22]
Operational manual of the monitoring system

D3.4 : Prototypes samples produced in the upscaled plant [18]
Prototypes samples produced in the upscaled plant

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS3	Upscaled design of MOGU facility	1 - MOGU	10	The design of the production plant will be available and described in D3.1
MS4	Building and commissioning of MOGU facility	1 - MOGU	18	Pilot plan in place and operative

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS5	Commercial production of MOGU floor achieved	1 - MOGU	18	MOGU floor available on the market
MS6	MOGU floor validated at TRL9 with stakeholders	1 - MOGU	24	MOGU floor installed in demonstration sites

Work package number [9]	WP4	**Lead beneficiary** [10]	1 - MOGU
Work package title	Certifications		
Start month	5	**End month**	24

Objectives

1) Obtain the CE mark for MOGU floor,
2) Obtain additional green certification for MOGU floor,
3) Obtain certification for the production process of MOGU floor

Description of work and role of partners

WP4 - Certifications [Months: 5-24]
MOGU
Task 4.1 CE mark
The application for receiving the CE mark will be filed in all its steps, analysing the prototypes produced in WP2, and submitted to one of the Italian accredited society (e.g..; Istituto Giordano S.p.a; Rina Services S.pa; CSI S.p.a). Task 4.2 Green certifications
The performances required to achieve the green certifications selected in WP1 will be preliminarily tested, according with the standard methods, in our laboratory if possible. Once the performance will be confirmed the application for the selected green certifications of MOGU floor will be submitted to the accredited bodies, according with the official procedures.
Task 4.3 Process certifications
The new plant for MOGU Floor production will be ISO 9001 certified. The adoption of the ISO certification system will ensure to adopt an effective and efficient system, along all the production line, not only in this developing phase but also, and especially when the plant will be fully operational. The Plan, Do, Check, Act cycle will be implemented through the process and will provide an instrument able to check MOGU performances and eventually re-think some aspects, to maintain the product compatible to the changing market needs.
Our target is to obtain also environmental certifications such as UNI EN 16810:2017 “Resilient, textile and laminate floor coverings - Environmental product declarations - Product category rules”.

Participation per Partner

Partner number and short name	**WP4 effort**
1 - MOGU	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type** [15]	**Dissemination level** [16]	**Due Date (in months)**[17]
D4.1	Report on MOGU floor certifications	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.2	Report on the manufacturing process certifications	1 - MOGU	Report	Confidential, only for members of the consortium (including	24

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type [15]	Dissemination level [16]	Due Date (in months)[17]
				the Commission Services)	

Description of deliverables

D4.1 Report on MOGU floor certifications [M18]
D4.2 Report on the manufacturing process certifications [M24]

D4.1 : Report on MOGU floor certifications [18]

Report on MOGU floor certifications

D4.2 : Report on the manufacturing process certifications [24]

Report on the manufacturing process certifications

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS7	Achieve CE mark	1 - MOGU	13	CE mark granted
MS8	Achieve additional green certifications	1 - MOGU	18	Green certifications granted
MS9	Achieve process certification	1 - MOGU	24	ISO certification granted

Work package number [9]	WP5	**Lead beneficiary** [10]	1 - MOGU
Work package title	Gear up for commercialisation		
Start month	1	**End month**	24

Objectives

1)Perform a technology watch activity and IPR management in order to protect our competitive advantage,
2) Perform an up-to-date Market Review and further develop our business plan to 'investor-ready' level,
3)Build up our commercialisation channels and distribution agreements for reaching international markets

Description of work and role of partners

WP5 - Gear up for commercialisation [Months: 1-24]
MOGU

Task 5.1 Technology Watch and Knowledge Management
Through this task technological developments and innovations will be identified and their impact for MOGU development will be analysed. In addition, up-to-date reviews of the state-of-the-art and patents will be performed periodically to reconfirm the novelty of the development and the freedom to commercialise. The results generated by the project will be assessed in terms of their inventive steps, novelty, patentability and protectability in line with development. An analysis of the most appropriate IPR tools to protect the different IP generated will be carried out and the related IPR tool filed accordingly.

Task 5.2 Market Segmentation and Business Planning
We will perform a market segmentation of the flooring market in order to plan and prioritise the segments that are most likely to pay for our product in the early market introduction phase, this way we can ensure we focus our marketing budget towards those segments that will give us the quickest returns and will position us to enter the growth phase. We will design appropriate Value Propositions tailored to each segment. We will review our SWOT analysis and further develop the business plan, working out from the Business Model Canvas approach towards an investor ready Business Plan-and associated pitch decks. The resulting business strategy will be used in the third stage of our fundraising strategy to further extend our sales in Asia and US in the mid-long term.

Task 5.3 Preparation for commercialisation
We will gear up to market by establishing the manufacturing infrastructure for being market ready within the completion of the Phase 2. Our Commercial Team will be enlarged and trained. Business development activities will be performed to build stakeholder networks and the supply chain, and to build business relationships to support long-term business collaboration. For promotional activities, MOGU, in collaboration with EXT, with whom we have a collaboration agreement, will host an annual workshop to coincide with the world famous Milan Design Week, an international interior design and furniture exhibition.
The commercial network will be established (agreements with first agents, distributors and licensees for key markets will be signed). As such, a database of distributors and licensees will be built, and maintained. The Commercial Team will engage with key contacts to generate sales revenue.

Task 5.4 Communication, dissemination and marketing
A communication plan will be elaborated and communication tools and materials will be developed (updated corporate website, video, leaflets, posters, press releases, articles for our newsletter, etc.). Our communication campaign will be implemented in line with this plan on several channels: websites and blogs, social media, magazines and so on. Publications will be prepared for submission to peer reviewed Open Access Journals, Technical Magazines, etc. The tradeshows, conferences and other events that we will be participating in during the 24 months as well as during the commercialisation phase will be planned and appropriate dissemination for the project will be implemented. During this task, we will also build and deploy the E-Commerce Portal that will be central to our digital sales channel. This Portal, which will be accessible from our corporate website, will allow us to showcase impactful images of our products, as well as provide product information, as well as pricing, as well as providing easy to use and secure tools for making online product orders.

Participation per Partner

Partner number and short name	WP5 effort
1 - MOGU	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	New MOGU website	1 - MOGU	Websites, patents filling, etc.	Public	3
D5.2	Communication Plan	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D5.3	Market Review	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D5.4	Report on status of patents or other IPR tools	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D5.5	Business Plan for the commercialisation phase	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D5.1 MOGU Corporate website overhaul [M3], with ecommerce portal (MS10) online [M18]
D5.2 Communication Plan [M6]
D5.3 Market Review [M9]
D5.4 Report on status of patents or other IPR tools [M18]
D5.5 Business Plan for the commercialisation phase [M24]

D5.1 : New MOGU website [3]

New MOGU website

D5.2 : Communication Plan [6]

Communication Plan

D5.3 : Market Review [9]

Market Review

D5.4 : Report on status of patents or other IPR tools [18]

Report on status of patents or other IPR tools

D5.5 : Business Plan for the commercialisation phase [24]

Business Plan for the commercialisation phase

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS10	Ecommerce portal	1 - MOGU	18	Ecommerce online
MS11	Business plan market ready and fundraising	1 - MOGU	24	Our business plan will be described in D5.5 and presented to investors
MS12	Agreements with stakeholders for early adoption	1 - MOGU	18	Contracts signed for demonstration of MOGU floor

Work package number [9]	WP6	**Lead beneficiary** [10]	1 - MOGU
Work package title	Management		
Start month	1	**End month**	24

Objectives

1) Monitor the work to ensure it is delivered on time, to budget and to the expected quality,
2) Ensure risk management procedures are implemented,
3) Manage the Innovation Process in a systematic manner,
4) Fulfil the contractual obligations of the Grant Agreement

Description of work and role of partners

WP6 - Management [Months: 1-24]
MOGU
Task 6.1 Monitoring and reporting
Ms Lisa Bigozzi will occupy the role of the Project Manager and will ensure project quality control by a process of self-assessment the application of a Quality Management System (QMS). Consistency with the workplan budget will be controlled. Fulfilment of the project schedule and milestones will be ensured through close will be monitored. Ms. Bigozzi will prepare the quarterly internal reports to be presented to the MOGU Management Board and will also ensure that all milestones are met and that all reports (including contractual deliverables) are prepared and delivered to the EC in accordance with contractual requirements.

Task 6.2 Risk Management
Risk Management procedures will be in place, whereby the status of the Critical Risks identified will be continually reviewed, and additional risks and mitigation strategies will be devised as appropriate. Contingency plans will be brought into play in an early and timely fashion.

Task 6.3 Innovation Management
This task will ensure an appropriate and systematic approach to Innovation Management is implemented. Federico Maria Grati will lead the Innovation Management activities. He will ensure that the innovation process is appropriately managed. He will lead a working session on Innovation activities that will take place every 8 months, with the presence of the Advisory Board.

Participation per Partner

Partner number and short name	WP6 effort
1 - MOGU	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type [15]	Dissemination level [16]	Due Date (in months)[17]
D6.1	Internal project management tool	1 - MOGU	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	1
D6.2	Compilation of Quarterly Reports to the	1 - MOGU	Report	Confidential, only for members of the	12

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type [15]	Dissemination level [16]	Due Date (in months)[17]
	Management Board for Year One			consortium (including the Commission Services)	
D6.3	Compilation of Quarterly Reports to the Management Board for Year Two	1 - MOGU	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D6.1 Internal project management tool [M1];
D6.2 Compilation of Quarterly Reports to the Management Board for Year One (Q1, Q2, Q3, Q4) [M12];
D6.3 Compilation of Quarterly Reports to the Management Board for Year Two (Q5, Q6, Q7, Q8) [M24]

D6.1 : Internal project management tool [1]

Internal project management tool

D6.2 : Compilation of Quarterly Reports to the Management Board for Year One

[12] Compilation of Quarterly Reports to the Management Board for Year One

D6.3 : Compilation of Quarterly Reports to the Management Board for Year Two

[24] Compilation of Quarterly Reports to the Management Board for Year Two

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP7	**Lead beneficiary** [10]	1 - MOGU
Work package title	Ethics requirements		
Start month	1	**End month**	24

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP7 - Ethics requirements [Months: 1-24]
MOGU
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type [15]	Dissemination level [16]	Due Date (in months)[17]
D7.1	H - Requirement No. 1	1 - MOGU	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2
D7.2	POPD - Requirement No. 2	1 - MOGU	Ethics	Confidential, only for members of the consortium (including the Commission Services)	2

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package. D7.1 : H - Requirement No. 1 [2]

- Details on the procedures and criteria that will be used to identify/recruit research participants must be submitted as a deliverable. - Detailed information on the informed consent procedures that will be implemented for the participation of humans (stakeholders, customers and end-users) and in regard to data processing must be submitted as a deliverable. - Templates of the informed consent forms and information sheets covering the voluntary participation and data protection issues (in language and terms intelligible to the participants) must be kept on file (to be specified in the grant agreement) and the English version must be submitted as a deliverable.

D7.2 : POPD - Requirement No. 2 [2]

- The beneficiary must confirm that it has appointed a Data Protection Officer (DPO) and the contact details of the DPO are made available to all data subjects involved in the research. For beneficiaries not required to appoint a DPO under the GDPR a detailed data protection policy for the project must be kept on file (to be specified in the grant agreement) and submitted to the Agency upon request. The confirmation for each beneficiary must be submitted as a deliverable. - The beneficiary must explain how all of the data they intend to process is relevant and limited to the purposes of the research project (in accordance with the ‘data minimisation ‘principle). This must be submitted as a deliverable.

	Schedule of relevant Milestones					
Milestone number[18]	**Milestone title**		**Lead beneficiary**		**Due Date (in months)**	**Means of verification**

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Design of the first MOGU floor collection	WP1	1 - MOGU	4	The first portfolio of MOGU floor will be presented: online, press release, events
MS2	Manufacturing protocol of MOGU floor	WP2	1 - MOGU	6	The full production protocol and conditions will be described in D2.1
MS3	Upscaled design of MOGU facility	WP3	1 - MOGU	10	The design of the production plant will be available and described in D3.1
MS4	Building and commissioning of MOGU facility	WP3	1 - MOGU	18	Pilot plan in place and operative
MS5	Commercial production of MOGU floor achieved	WP3	1 - MOGU	18	MOGU floor available on the market
MS6	MOGU floor validated at TRL9 with stakeholders	WP3	1 - MOGU	24	MOGU floor installed in demonstration sites
MS7	Achieve CE mark	WP4	1 - MOGU	13	CE mark granted
MS8	Achieve additional green certifications	WP4	1 - MOGU	18	Green certifications granted
MS9	Achieve process certification	WP4	1 - MOGU	24	ISO certification granted
MS10	Ecommerce portal	WP5	1 - MOGU	18	Ecommerce online
MS11	Business plan market ready and fundraising	WP5	1 - MOGU	24	Our business plan will be described in D5.5 and presented to investors
MS12	Agreements with stakeholders for early adoption	WP5	1 - MOGU	18	Contracts signed for demonstration of MOGU floor

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	LOW: low appreciation of MOGU collection	WP1	Consumer panels and co-design with stakeholders
2	MEDIUM: low efficiency of manufacturing process	WP3	To develop a topnotch production plant, in line with the principles of Industry 4.0, the support of IRIS, an advanced engineering company, will be subcontracted (see section 4)
3	LOW: failed achievement of CE mark	WP4	MOGU floor has been preliminary tested with positive results
4	LOW: failed achievement of green certifications	WP4	A wide range of labels will be screened to select the most suitable ones
5	MEDIUM: low awareness of MOGU floor benefits	WP5	Marketing campaign is designed with promotional installations
6	MEDIUM: limited distribution channels	WP5	Preliminary agreement signed at international level and ecommerce portal will be prepared

1.3.6. WT6 Summary of project effort in person-months

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total Person/Months per Participant
1 - MOGU								
Total Person/Months								

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	**Tentative timing**	**Planned venue of review**	**Comments, if any**
RV1	12	Remote	
RV2	24	Remote	

4. Members of the Consortium

4.1 Participants

MOGU is a start-up company based in Inarzo (VA – Italy) that focuses on the use o f mycelium-based technologies to develop a family of s ustainable bio-composite materials, following the principles of the Circular Economy. This material, composed by mycelium – a bio-polymer constituting the root-structure of mushrooms – and by-products of agro-industrial value chains, is produced through a natur al process and it is 100% compostable. MOGU is suitable for applications in diverse markets, with a special focus on housing materials for green-building and bio-design.

Since its foundation, MOGU has be en very active in the technical development (m aterials and final products), testing and optimizing sev eral growth protocols, implementing different fee dstock and fungal strains, and assessing final product immprovements (e.g. cold and hot pressing, coating application, extra drying, etc.). Such activities have been undertaken in the own and well equipped industrial laboratory (about 300 sqm), and in collaboration with ke y partners, such as final users, feedstock suppliers (from agricultural and industrial sectors), as well as pri vate and public Research Institutions (e.g. Utrecht University, Pavia University), with which important a greements have been concluded. At the curren t time, MOGU is developing a range of different produc ts in the identified market segments. Thanks to th e continuous R&D activities, MOGU has accumulated a si gnificant technical and market-related know-how, b ringing the MOGU brand to the attention of both industria l and public stakeholders.

MOGU has planned its market entry i n 2018, selling sustainable bio-panels and tiles for the green building and bio-architecture sectors; such activ ity is carried out in collaboration with a European partner providing a finishing for MOGU products. The Com pany's market strategy is focused on the commercialization of final products (B2C) in the short term (2017- 2018), entering the licensing phase in the medium term (2021).

During the first 3 years The Company has been funded by the shareholders (120 k€), and granted by the European Community under the Horizon 2020 (about 450 k€ - 3 projects 2016-202 1). The cumulated turnover (2017-2018) is about 150 k€ (80% consulting activities, 20% product sales). On December 2017, the Milan based Investment Fund Moffu Labs Srl acquired the 5% of MOGU shares, valida ting a post money value of 1,7 M€.

MOGU Team is actually composed by 10 FTE, showing a consistent growth, with a forecast of more than 50 new hiring during the development of the project. According to the Company core business, the Team has a multidisciplinary composition (engineers, microbiologists, designers, and market specialist s), covering all the fundamental expertise.

MOGU is Member of Lombardy Green Chemistry Association (https://www.chimicaverdelombardia.it/en/), and it is also part of Assobiotec (Federchimica - Confindustria Italia) (http://assobiotec.federchimica.it) Team

Stefano Babbini CEO

Stefano has been active in Bio-economy and agro-Industry sector since 2 005. Graduated in Environmental E ngineering in Florence University, he contributed to start-up and ramp-up of relevant initiatives in EU and in Developing Countries, in the bio-en ergy, agro-industry and green chemi stry (about 100 stable jobs created). In the past he co-fu ndedAgroils (2007-2011 - a consulti ng and developing company in the biofuel sector), Treedom (2010-2012 - a reforestation com pany offering solutions in the CO2 compensation), Smart Oil Ltd (2008 – 2016 - an agro-c ompany in West-Africa). Beside the management experience developed in field, he increased his skills attending exclusive programs focused in the Business & Administration (U nicreditStartlab Academy, Alimenta2Talent). He co-fun dedMOGU in 2015, and today he is fully dedicated to its business development, in charg e of the Company business plan and strategic decisions. He is also responsible of MOGU partnership in BBI Horizon 2020 Con sortia.

Maurizio Montalti R&D Director

Maurizio is a designer, engineer and creative entrepreneur with 10 years' experience with production of m ycelium materials. Maurizio operates as a trans-dis ciplinary designer, mainly focused on researching and developing ways for implementing biological organisms as materials and consumer products, undertaking its research in collaboration with Utrecht University. His w ork has been honored multiple awards and it has been extensively shown in museums, gall eries and festivals. Co-founder and shareholder of MOG , he is responsible for leading R&D a ctivities and hands-on experimentation in MOGU.

Federico Grati

BoD Member

Federico is an en vironmental engineer with 10+ ars'experience in the bioenergy and bio-material sector. Collaborating with Stefano since 2006, he co-founded various innovative start-ups (AgroilsSrl, TreedomSrl and Smartoil Ltd. in Ghana) and has been honored with BEST Fulbright scholarship and GSBI in 2011. Since 2013, he has worked on advanced biofuels in the chemical industry (Mossi&Ghisolfi, Clariant, and Ferrero). Experience in relationship management with EU for Horizon2020, Red Directive, and Sustainability. Co-founder and shareholder of MOGU, Federico is Board Member and Business Advisor, with a special focus on the Company strategy.

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Antoni Gandia CTO

With a Master in Plant Molecular Biology (IBMCP), he has been working in the technology development of bio-materials from microorganism, with a special focus on mycelium bio-composites (expe rience in the US company Ecovative Design). During the last 5 years Antoni tested several growth protocols with different fungal strains for ma terials production, collaborating als o in the mechanical post-treatments set-up. He will b e in charge of the technical develop ment of the company, from the processes to the products characteristics.

Serena Camere

Prod.Dev.Manager

Serena is a multidisciplinary industrial designer and Post-Doc researcher working with bio-based Emerging Materials. During her studies, she collaborated with several design studios and industrial co mpanies, such as Skitch, Serralunga, Alessi, Woodnotes, Camparisoda, Eurochocolate, Design Innovation and Fiat-Chrysler. She is in charge of products development, according to technical and design requirements, and reporting to CTO and CEO.

Lisa Bigozzi

Process Manager

Lisa is an Environmental Engineer, with Master in bio-energies (Imes - Florence). She has been worki ng (5 years) in the management and supervision of building projects. She is also experienced in EU-funds projects (ETA Florence). Lisa is t aking care of the industrial scale- up of the MOGU process, and the products development. She is also supporting the m anagement of strategic research partnerships.

Alessandro Forzinetti

Commercial Director

Alessandro is a graduate in economics (Insubria University Varese).S enior commercial manager with le ngthy experience in several sectors (building materials, telecom, energy). He is responsible for designing and implementing MOGU's commercial strategy in the short and long term, as well as guiding the product development effort in line w ith market needs.

Francesco Amadio

Process Manager

Francesco is an environmental engineer with 10 years' experience in the renewable energy sector, energy efficiency and saving, quality and auditing, product and processes labelling and certification. In 2016, he completed a II Level Master Course in Environment and Innovation Management at LUISS Business School in Rome. Co-founder and shareholder of MOGU, he is resp onsible for all technical aspects and certification tasks.

Francesco Giannetti

Product Manager

Francesco is an industrial designer and architect with outstanding creativity and technical skills in object d esign, 3D graphic. Thanks to his experience in furniture design, he is supporting new and customized MOGU applications in different fields.

Olga Volkova **Market Specialist**	**Olga is a market specialist and project mana**Associated **with**with document**10years**Ref**experience**.Ares(2018)3790472**isthe** - 17/07/2018 **development of commercial activities in the interior design and building sector in Baltic Countries, Germany and Sweden. She is in charge of the market strategy in North EU.**
Mirko Pellegrini **Biotechnogist**	**Graduated in Insubria University (Varese) in 2016. He has significant laboratory experience, mostly working in the characterization of enzymes activities, isolated from metagenomic libraries. In MOGU, he is involved in the tasks related to the fungal growth in different substrates.**
Marco Cartabia **Mycologist**	**Thanks to his great passion for fungi and to his university studies, Marco has developed a valuable know-how in the fields of mycelium and mushrooms. After his experience at EMPA (Material Science and Technology, St. Gallen CH) and after his graduation in Agricultural Science in University of Milan (master thesis about biological control of wood decay fungi), he joined MOGU team in Jan 2017, assisting with daily lab activities.**
Monica Ghiringhelli **Back office and commercial interface**	**Monica has a long experience in the office management, coordinating the accounting and administration activities. She is also supporting the Company in the 'front of house' interface with customers and suppliers.**

Relevant publications/products/services related to project successfully accomplished

Press articles **MOGU has appeared, and has been interviewed, in several national and international media such as Il Sole 24 Ore (Food 24), Canale 5 (Striscia la notizia), Corriere della Sera, Il Giorno, Renewable Matter, Le Scienze, La Prealpina, Ansa, Reteconomy, Rai 2 (2Next), TV2000, Linkiesta, Ecopreneur Blog.**

► ANSA – Young Italians Invent Biomaterial(2015)
►Il Sole 24 Ore –È italiana la bioplastica fatta con i funghi e gli scarti agricoli [(2)] (2015)
►Corriere della Sera –Nasce l'imballaggio fatto di funghi (2015)
► Renewable Matter – Distributed generation finally hits bioplastics production (2016) ►Il Giorno – Cascine di Milano e start up tecnologiche, al via il progetto 'OpenAgri'(2017)
►Il Bioeconomista–A startup produces biomaterials from fungi. An interview with its co-founder Federico Grati (2017)
►Canale 5 - Striscia la Notizia – Occhio al futuro (2017)
► Greennews.info - Pannelli e pavimentazioni per la bioedilizia a partiredaifunghi (2017)

Presence at ► 2018 – Klimahouse – Bolzen (Italy) – Presentation and products exposition Exhibitions, ► 2017 – Biofabricate – NYC (USA) – Presentation and products exposition Tradeshows ► 2017 – EFIB – Brussels – European Forum for Industrial Biotechnology - Presentation and ► 2017 – World Responsible Economy Forum – Arras (F) - Presentation Conferences ► 2017 – Ecomondo – Rimini (I) – Bio-economy best national cases - Presentation

► 2016 – EIP AGRI workshop on biobased economy, Oristano, (I) – Participation
► 2016 – Biobased start-up day (Nova-Institute) [(X)], Cologne (D) – Presentation
► 2015 – Italian Forum Industrial Biotechnology (IFIB) [(X)], Lodi, (I) – Presentation

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BBI GRACE BBI project 01/06/17 31/05/22 (2017-2022) Growing Advanced Industrial Crops on marginal lands for bio-refineries. Call H2020-BBI-2016-D2 - Improvement and adaptation of industrial crop varieties and novel sources of biomass to diversify biomass feedstock for biorefineries. Budget / Grant MOGU: €337,500 / €236,250 - MOGU contribution: MOGU will demonstrate the opportunity of valorizing hemp and miscanthus fibers producing bio-panels and bio-tiles using mycelium technology.

BBI AGRIMAX BBI project 01/10/16 30/09/20 (2016-2020) Agri and food waste valorization co-ops based on flexible multifeedstocks biorefinery processing technologies for new high added value applications. Call BBI.VC3.D5-2015 - Valorisation of agricultural residues and side streams from the agro-food industry refinery processing technologies for new high added value applications-Budget / Grant MOGU: €267,500 / €187,250-MOGU contribution: MOGUis project leader on the task related to the use of bio-waste and derived compounds in bio-polimers, with a special focus on the production of mycelium based materials from bran. To this end, MOGU will test the mycelium colonized materials to produce panels, gardening and packaging applications.

NWO Traffic control; optimal flow of nutrients to developing mushrooms – NWO – Utrecht University is leading a research finalized to the optimization of nutrients usage for mycelium and mushroom production. Total project budget: €821,580. MOGU is a member of the User Committee, contributing to the project tasks for about €12,320, involved in the prototypes production and product characterization.

STW Mycelium based material for product design. STW – Program: Research Through Design. The project aims at the development of a palette of mycelium-based composite materials. These compostable materials will be used to create, among others, plant pots and a cupboard, that will serve as a model for other applications in product design. Total project budget: €437,000. MOGU is a member of User Committee, contributing to the project tasks for about €25,000, involved in the prototypes production and product characterization.

Prizes Since its foundation, MOGU Team has been very active in marketing activities, in multiple events, such as conferences, tradeshows, thematic workshops, but also startup competitions, with the aim of marketing the MOGUcompany and products to relevant stakeholders of differing backgrounds (financial, technological, academic, etc.). MOGU concept has been very well received in several environments, encouraging the business development during the first year.

MOGU has been honoured with the following awards:

►2018 – MOGU is finalist at KlimahouseStartupAward (http://www.fierabolzano.it/klimahouse/startup.htm)

►2016 – MOGU won the "Alimenta 2 Talent" competition at Seeds and Chips 2016. Champion of the 3rd Edition of Alimenta 2 Talent Competition for Circular Economy Start Up of the year. (http://www.alimenta2talent.eu)

► 2016 - Unicredit Start Lab Academy (2 month) in Milan, with the aim of set up the best company strategy for the project scale up.-

►Finalist at BioInItaly Investment Forum 2015 (http://assobiotec.federchimica.it)

► 2015 - Best Idea in The Green and Circular Economy at Global Social Venture Competition 2015, promoted by Startup Initiative (Banca Intesa) in Milan

Maurizio Montalti as Mycelium Designer

► June 2016 finalist for New Material Award
► 2016 (NL)-DOEN Grant (2015-2016). The Growing Lab - Mycelia
► (2015). Manchester Metropolitan University acquiring part of "The Growing Lab". Multi-disciplinary research project on the use of fungal mycelium as a material for applications in design, architecture, fashion and beyond.
► MBAM Collections (2015). Montreal Museum of Fine Arts acquiring part of "The Growing Lab"
► Fungal Futures (2016): Growing Domestic Bio-landscapes, presents the work of an international group of artists and designers who develop innovative materials and applications using mycelium, the rhizomic network of 'fungal roots' (Maurizio Montalti – MOGU Partner).
► The Future of Plastic (2014): commissioned by PLART Foundation, Naples Project, Exhibition and Symposium on the implementation of fungal/mycelium materials as potential replacement for traditional synthetic plastics.
► Fungi – The Plastic of the Future: VICE documentary featuring the work of OfficinaCorpuscoli/Maurizio Montalti (MOGU partner).

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Pilot Plant MOGU has implemented an own pilot plant (300 sqm) in Inarzo (VA), increasing its Varese, Italy prototyping capability with daily production capacity of about 120 kg of biomaterials. Suitable machineries have been installed, such as steam autoclave, incubation cabin, biomass mixer, methane heating plant, ventilation system, biomass grinder, molds and plates and a mechanical press. The activities are conducted into a controlled environment, keeping the temperature constant and monitoring the air quality through a ventilation system. Colonized substrates are growing into dedicated industrial shelfs. A clean area, equipped with laminar flow and EPA filters, is used for fungal strain inoculation. The facility includes also an electric oven for material drying, a storing space, handling tools, and finally a comfortable office and meeting room.

The mechanical pressing for the tiles and panels production is performed by a suitable machine (electric hydraulic press), able to compress the composite materials at cold and hot temperature.

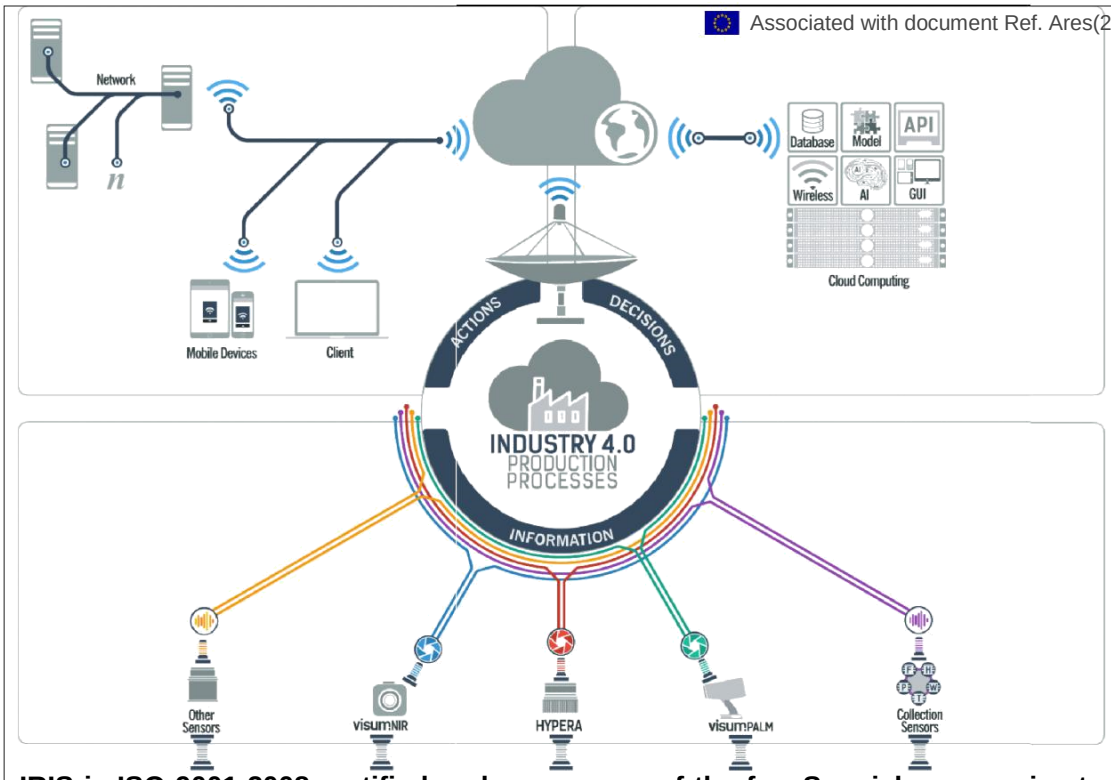
Growing and molding tests are also conducted in the Maurizio Montalti workshop in Amsterdam, where small-scale samples can be assessed.

The team uses the Microsoft software and Rhinoceros for 3D design. Mould prototyping is conducted in collaboration with partners in Italy and Amsterdam. For very specific activities related to microbiological aspects of the materials, MOGU has established a collaboration agreement with Utrecht University and Pavia University (and its spin-off MiconetSrl).

BBI Horizon 2020

Thanks to the participation to two BBI projects, MOGU improved its pilot facility, with the purpose of implementing a demo facility for the manufacturing of MOGU products. In particular, the demo plant includes the three key sections: substrate production line (sterilization system of about 100 kg/day), a multi-layer mechanical press (up to 30 sqm/day) and a drying equipment (conditioned cabin). Storage facility will be upgraded as well, installing an industrial shelf system for feedstock and semi-finished products warehousing.

Do you plan to subcontract any tasks?	Yes
MOGU will subcontractsome of specific tasksinWork Package 3- Production upscale, and in particular, in relation to the design, development and commissioning of a monitoring and control system for operating the production process efficiently at larger scale and handle the increased mass fluxes predictive model of the process aimed at supporting the decision making process and simulating new products manufacturing based on new raw materials and parameters. While MOGU is expert in the manufacturing process at biotechnological and engineering level, we do not have the expertise necessary for the development of the processing monitoring and control system, as this requires specialised knowledge and skills in sensor technology, optical engineering, predictive modelling, among others that MOGU does not possess in house, and which are not core competences that are necessary for scaling the actual business itself. **As such MOGU will subcontract this work to an advanced engineering company from Spain, IRIS Technology Solutions. Hereunder we provide a description of their core expertise, and the team that will be involved in performing the work, which highlights their suitability to perform the role. The novelty of MOGU technology requires the development of specific tailor-made solutions to meet our manufacturing needs. In the past we enquired and collaborated with several engineering companies, in Italy and Europe, to solve our technology needs. The best results so far have been achieved with IRIS that combining its NIR based monitoring device with a customized chemometrics model has been able to measure the key parameters in the inoculum phase of the process. In short, in view of their expertise in spectroscopy analysers, optical engineering, sensor integration, chemometric modelling, mathematical modelling and data processing, software development, cloud computing, as well as their proximity to us and the fluidity of our working relationship to date (whereby there has been a seamless engagement with the IRIS engineers in the initial conceptualisation of the inline monitoring system for our manufacturing line). We already have a good working relationship with IRIS, we have a long standing NDA, and we have an agreement in place that ensures that all Results from their work in this project will be owned by MOGU. As such, therefore, in virtue of the consolidated knowledge about our process and the specific monitoring technology the subcontracting of IRIS represents the best value for money.** **IRIS (www.iris-eng.com) is a technology SME that was established in Barcelona in 2007 and specialises in photonics (spectroscopy- predominantly Near Infrared Spectroscopy, Hyperspectral Imaging and Raman) and sensor technology integration and SMAC technology (Social, Mobile, Analytics and Cloud) for the development of advanced monitoring and control solutions, and for IoT-enablement for transitioning industry to the Industry4.0 paradigm (the image below provides a schematic overview of IRIS positioning as Key Enabling Technology provider in the Industry 4.0 context).**	



IRIS is ISO 9001:2008 certified and we are one of the few Spanish companies to also hav e UNE-EN-ISO 166002: Management Systems R&D&I certification.

Expertise and capacity to perform the work

In relation to experience, IRIS has dev eloped and integrated numerous inline systems in food processing, pharmaceutical, chemical and OEM applications, based on NIR and hyperspectral imaging solutions for foreign body det ection and chemical composition analysis, as well a s data management solutions for data processing, decision support, automation and data analysis (big production data).

IRIS currently employs ~60 Full Time Employees highly specialised staff, and its team in cludes telecom and electronics engineers, system engineering, application programming and web development etc. IRIS can carry out full system design and engineering, including electronic design (PCB) and assembly, firmwar e and software development, electrical harness es and interconnections. Moreover IRIS has o ptical engineers, mechanical and mechatronics engineers, specialised in process upscaling, integration work, as well as installation, validation and industrialisation of prototypes and eq uipment in different industries. In addition to this broad cross-disciplinary engineering team, IR IS' research team consisting among others of fo od and material scientists, biologists and physicists, have all the expertise to validate the techn ologies developed. Finally, IRIS has significant expertise in the use of QFD (quality function deplo yment) and Kano, and uses a systems develo pment management approach based on KANBA N agile methodologies.

In relation to equipment, IRIS has a Spectroscopic Laboratory: equipped with several instruments, namely, an FTIR-ATR (Inte rspectrum), two UV-VIS (PerkinElmer and Ocean O ptics), one Extended-SWIR (StellarNet) and one NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenic s). The associated staff are trained and skilled in chemometrics modelling by using a co mmercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes). Electronics, mec hanical and optical workshops, production/ass embly facilities: equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electron ic and

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mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the design of optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

The key personnel who will be involved in the development work

Dr. Ing. ELODIE BUGNICOURT (female). Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Leader of the Eco-Materials business unit in IRIS since 2009. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: R&D in chemical industry especially related to reactive polymer formulations: i) Arkema (ex. Cray Valley, 2002, Spain) in the field of polyester powder coating; Wacker Chemie (2002-2005, Germany and France), Silicone division that funded her PhD thesis: development of nanocomposites based on an epoxy matrix and silica; iii) Ashland (2006-2009, located sequentially in US, Spain and France) in the Composite Polymers division, focussing on bio-sourced resins, flame retardant resins, SMC resins, PU and epoxy structural adhesives. Led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical methods and modelling (for improved quality management); as well as collaborations with academic labs and other players in the supply and value chain. Author of several papers in peer-reviewed and technical journals, and has presented her work in numerous international conferences. She will lead the development team in IRIS and engage with the client, SNELL Optics.

DR. TIMOTHY KEHOE (MALE) PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterision of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials.Tim is the Technical Manager of the Optics Group, and he brings vast optical integration and device instrumentation expertise.

DR. SEVERINE PHILIPPE (FEMALE) graduated with a 1st class Hons. B.Sc. in Applied Sciences (Physics and Physics Technology) from Dublin Institute of Technology, Ireland, in 2002 and received her PhD in Physics from Trinity College Dublin in 2007, for the study of polarisation effects in semiconductor optical amplifiers (SOA). She went on to work on the development and characterization of low noise SOAs, as well as THz generation by multimode lasers and their detection using four wave mixing in SOA, both in the Semiconductor Photonics Group, Trinity College Dublin, and the School of Electronic Engineering, Dublin City University. She previously worked as a R&D technician for the plastics industry and in quality control for electronic appliances. She joined IRIS in 2014 as an optical engineer and is currently involved in projects including various microscopy techniques, planar waveguide technologies and MIR spectroscopy.

DAVID MONTES MONTESERIN. Male. Technical Manager. BSc in Mechanical Engineering (2011), Polytechnic University of Catalonia (UPC), Barcelona. Prior to joining the Production Processes group at IRIS as Mechanical Design Engineer, he gained wide experience in product design and development, including dosing systems for the chemical industry and both industrial and domestic appliances for the water treatment sector. He has also held positions as Engineering Team Manager, applying renewable energies to industrial-sized drinking water plants among other projects. He is specialised in design for manufacturing and is proficient in SolidWorks, design software which has taught to several companies. During this project, he will lead the design and building of the system hardware and will work closely with BIOTRONICS and DENKA. He is the technical manager of the engineering team in IRIS and he has a team of mechanical,

electronic and mechatronic engineers that will work on the PCD development.

JOAN A. MARTÍNEZ BORRÀS. Male. Telecommunications Engineer from the Universitat Polytechnic de Catalunya specialising in Electronic Systems. Broad experience in the development of embedded systems working with microcontrollers from Intel, Nec, Motorola, National Semiconductor, Zilog, Renesas (Hitachi) and Atmel and its related software written in C and assembler. Expertise in R&D and industrial development- Indomatic Elite, Dam Alignment By Means Of Laser for Endesa and test devices (control of the electropneumatic test device for the SEAT Toledo dashboard manufactured by VDO). His communications libraries following the V.21, V22, V.22bis and X.25 standards for Electronic Funds Transfer terminals were deployed on a few million payment terminals around the world and his teleloading system allowed some hundreds of thousands of terminals to collect the transactions and update their software online. He also has led the development of some security EFT terminals following the EMV, ZKA, VISA PED and PCI PED standards. His programming skills include assembler, C, C++, Java, Visual Studio, .NET, Linux, Qt, Gtk, HTML, CSS, Javascript, PHP. Joan will lead the data communication and integration work in the development.

IVAN PAQUICO RODRIGUEZ. Male. Scrum Master. Telecommunications Degree in UPC (Barcelona) and a Joint Degree in Multimedia Development in LUAS (Finland). His final thesis, 'Augmented Reality Chess and Application to Detect Chess Moves', implemented by using Computer Vision, Augmented Reality and Artificial Intelligence techniques, has won several awards, such as the best final thesis of the UPC and the best final thesis of Spain (received from Vodafone Spain and Universia foundation). He is an official iOS and Android Developer since 2010, creating a social network for iPhone called: 'BeThere' developed by iOS SDK from Apple and used Couchbase technology for the database. He also has experience in different programming languages such as C, C++, Objective C or Java, 8 bits embedded and real time systems. He has expertise in multimedia methodologies ranging from Virtual Reality programming techniques to 3D/2D modelling software such as OpenGL, Quest3D, Autodesk Maya, Autodesk 3DS Max or Adobe Photoshop. Ivan will lead the Scrum Team of software developers in IRIS who will work on the software development for building the cloud database, APIs, and setting up the cloud hosting. He leads a team of 10 programmers/coders.

JOSE ANTONIO FUSTE. SW Developer. PhD. in Signal Theory and Communications, UniversitatPolitècnica de Catalunya, Spain (2014), Thesis title: "Design Rules and Optimization of Electro-Optics Modulators Based on Coplanar Waveguides"; the project dealt with the development of new models to design high speed electro-optical modulators based in LiNbO3, where a novel software to design and optimize these modulators was created. Master Degree in Telecommunication Systems from Universidad de Oriente, Cuba (2005).Telecommunication Engineer Degree from Universidad de Oriente, Cuba (2003). Author of several papers in journals of high international impact. Experienced in optical communication softwares as Virtual Photonic (VPI), OptSim, and in electromagnetic structures simulators as High Frequency Structural Simulator (HFSS) and ADS. Experience for more than 5 years in programming languages as C++, C, Python, Matlab and SQL. He is currently working as a SW developer in IRIS, in the development of cloud platforms with technologies based on Python's frameworks as Falcon, mySQL database design, ORM Python SQLAlchemy, HTML, etc. He has expertise working on internal, commercial and collaborative R&D projects in the development of software for advanced monitoring systems, Cloud-based ERP systems, among others.

MSc. ALEJANDRO ROSALES. Male. M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS-NIR-MIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. His career has included positions as researcher applications engineer, design engineer, Business Developer, commercial and R&D Manager. In 2000s, he worked in the FTIR spectroscopy field

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applied to quality control in the wine industry, whereby he developed two new measurement instruments for wine analysis. His career as industrial physicist has been focused in applying his wide expertise to provide solutions for industrial problems. Knowledge in PAT, QbD, DoE and he has delivered numerous courses to industry in the field of practical PAT adoption. Knowledge in leading the building of chemometrics and multivariate analysis modelling tools.	
Will any of your linked third parties work in the action tasks?	**No**
Will you use contributions in kind provided by third parties?	**No**

5.1 Ethics

No ethic issues apply to this proposal

5.2 Security

- **Activities or results raising security issues: NO**
- **'EU-classified information' as background or results: NO**

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